

M190 Heart Failure - Integrated Admission & Discharge Rule Hierarchy

Admission Discharge. Domain Level-3 paths are shown with linked definitions embedded under each rule.

Definition

Composite Definitions

- [altered_mental_status_condition_present](#)
- [altered_mental_status_severe_or_persistent_condition_present](#)
- [cardiac_arrhythmia_of_immediate_concern_condition_present](#)
- [hemodynamic_instability_condition_present](#)
- [hypotension_condition_present](#)
- [hypoxemia_condition_present](#)
- [orthostatic_hypotension_condition_present](#)
- [persistent_tachypnea_despite_observation_care_condition_present](#)
- [respiratory_abnormality_condition_present](#)
- [severe_anasarca_or_peripheral_edema_condition_present](#)
- [tachycardia_condition_present](#)

Atomic Numeric Definitions

- [arterial_or_venous_ph](#)
- [arterial_po2_mmhg](#)
- [lactate_mmol_l](#)
- [mean_arterial_pressure_mmhg](#)
- [orthostatic_diastolic_blood_pressure_drop_mmhg](#)
- [orthostatic_systolic_blood_pressure_drop_mmhg](#)
- [shock_index](#)
- [spo2_percent](#)

Atomic Boolean / Leaf Definitions

- [altered_mental_status_severe_persistent_clause_condition_present](#)
- [anasarca_definition_generalized_edema_condition_present](#)
- [automatic_icd_firing_malfunction_or_setting_adjustment_needed_condition_present](#)
- [baseline_sbp_decrease_with_symptoms_condition_present](#)
- [clinically_active_atrial_fibrillation_condition_present](#)
- [coma_condition_present](#)
- [confusional_state_condition_present](#)
- [continuous_long_term_ecg_monitoring_needed_condition_present](#)
- [electrolyte_abnormality_not_at_acceptable_baseline_condition_present](#)
- [end_organ_dysfunction_due_to_hypotension_condition_present](#)
- [increased_supplemental_oxygen_from_baseline_condition_present](#)
- [lethargy_condition_present](#)
- [obtundation_condition_present](#)
- [persistence_despite_appropriate_treatment_context_present](#)
- [severe_peripheral_edema_manifestation_clause_condition_present](#)
- [stupor_condition_present](#)
- [supplemental_oxygen_required_to_maintain_targets_condition_present](#)
- [suspected_cardiac_ischemia_as_cause_or_consequence_of_vt_condition_present](#)
- [tachypnea_persistence_despite_observation_care_condition_present](#)
- [vasopressor_or_inotrope_required](#)

Needs Review / Pediatric Placeholders

- [acute_myocardial_ischemia_with_heart_failure_admission_condition_present](#)

- acute_myocardial_ischemia_with_heart_failure_admission_condition_present_verbatim_fact_condition_present
- acute_renal_insufficiency_evidence_gt_50pct_egfr_reduction_from_baseline_condition_present
- acute_renal_insufficiency_evidence_gt_50pct_egfr_reduction_from_baseline_condition_present_verbatim_fact_condition_present
- heart_rate_bpm
- impaired_airway_protection_condition_present
- pediatric_hypoperfusion_due_to_hypotension_condition_present
- pediatric_shock_index_threshold_condition_present
- pediatric_systolic_blood_pressure_threshold_condition_present
- pediatric_vasopressor_or_inotrope_required
- persistent_dyspnea_above_baseline_admission_clause_condition_present
- persistent_dyspnea_above_baseline_despite_observation_care_condition_present
- progressive_renal_insufficiency_evidence_gt_25pct_egfr_reduction_condition_present
- progressive_renal_insufficiency_evidence_gt_25pct_egfr_reduction_condition_present_verbatim_fact_condition_present
- pulmonary_artery_catheter_monitoring_needed_condition_present
- pulmonary_artery_catheter_monitoring_needed_condition_present_verbatim_fact_condition_present
- pulmonary_edema_insufficient_improvement_with_observation_care_diuretics_condition_present
- pulmonary_edema_new_or_increased_oxygen_need_for_spo2_target_condition_present
- pulmonary_edema_with_oxygen_need_and_observation_failure_condition_present
- respiratory_rate_bpm
- severe_electrolyte_abnormality_requiring_inpatient_care_condition_present
- severe_electrolyte_lab_line_candidate_condition_present
- systolic_blood_pressure_mmhg
- very_severe_pulmonary_edema_condition_present
- very_severe_pulmonary_edema_condition_present_verbatim_fact_condition_present

M190 — Admission

Admission is indicated for 1 or more of the following

Path 1: Hemodynamic instability)))))

- [atomic / IS_TRUE / hemodynamic_instability_condition_present] Hemodynamic instability (1) (2) (3) (7) (8) (9)

Linked shared definition: hemodynamic_instability_condition_present -> comp.hemodynamic_instability.root

Source: Definition - Hemodynamic instability

- **OR hemodynamic_instability_condition_present**
Hemodynamic instability, as indicated by 1 or more of the following
 - **OR vital_sign_abnormality_not_corrected_condition_present**
Vital sign abnormality not corrected by appropriate treatment, as indicated by 1 or more of the following:
 - **AND persistent_tachycardia_condition_present**
Tachycardia that persists despite appropriate treatment (eg, treatment of underlying cause, despite observation care)
 - **OR tachycardia_condition_present**
Tachycardia,[A][B] as indicated by 1 or more of the following:
 - **heart_rate > 100 bpm**
Heart rate greater than 100 beats per minute in adult[A][B](1)
 - **heart_rate > 85 bpm (needs_review)**
Heart rate greater than 85 beats per minute in child 13 to 17 years of age[A][C](2)
 - **heart_rate > 95 bpm (needs_review)**
Heart rate greater than 95 beats per minute in child 6 to 12 years of age[A][C](2)
 - **heart_rate > 110 bpm (needs_review)**
Heart rate greater than 110 beats per minute in child 1 to 5 years of age[A][C](2)
 - **heart_rate > 120 bpm (needs_review)**
Heart rate greater than 120 beats per minute in infant 3 to 11 months of age[A][C](2)
 - **heart_rate > 150 bpm (needs_review)**
Heart rate greater than 150 beats per minute in infant 1 or 2 months of age[A][C](2)
 - **persistence_despite_appropriate_treatment_context_present**
persists despite appropriate treatment
 - **AND persistent_hypotension_condition_present**
Hypotension that persists despite appropriate treatment (eg, treatment of underlying cause, despite observation care)
 - **OR hypotension_condition_present**
Hypotension, as indicated by 1 or more of the following:
 - **OR hypotension_in_adult_patient_condition_present**
Hypotension in adult patient, as indicated by 1 or more of the following:
 - **SBP < 90 mmHg**
Systolic blood pressure less than 90 mm Hg[A][B](1)
 - **MAP < 70 mmHg**
Mean arterial pressure[C] less than 70 mm Hg[A][B] MAP Calculator(1)(2)
 - **baseline_sbp_decrease_with_symptoms_condition_present**
Decrease in baseline systolic blood pressure of 30 mm Hg or more, with significant signs or symptoms due to lower blood pressure (eg, near syncope, syncope, chest pain)[A][B](1)
 - **OR hypotension_in_pediatric_patient_condition_present**
Hypotension in pediatric patient, as indicated by 1 or more of the following:
 - **SBP < 110 mmHg (needs_review)**
Systolic blood pressure less than 110 mm Hg in child 13 to 17 years of age[A][D](3)
 - **SBP < 100 mmHg (needs_review)**
Systolic blood pressure less than 100 mm Hg in child 6 to 12 years of age[A][D](3)
 - **SBP < 95 mmHg (needs_review)**
Systolic blood pressure less than 95 mm Hg in child 3 to 5 years of age[A][D](3)
 - **SBP < 90 mmHg (needs_review)**
Systolic blood pressure less than 90 mm Hg in child 1 or 2 years of age[A][D](3)
 - **SBP < 80 mmHg (needs_review)**
Systolic blood pressure less than 80 mm Hg in infant 6 to 11 months of age[A][D](3)
 - **SBP < 70 mmHg (needs_review)**
Systolic blood pressure less than 70 mm Hg in infant 3 to 5 months of age[A][D](3)
 - **SBP < 65 mmHg (needs_review)**
Systolic blood pressure less than 65 mm Hg in infant 1 or 2 months of age[A][D](3)
 - **persistence_despite_appropriate_treatment_context_present**
persists despite appropriate treatment
 - **AND persistent_orthostatic_hypotension_condition_present**
Orthostatic hypotension that persists despite appropriate treatment (eg, treatment of underlying cause, despite observation care)

- **OR orthostatic_hypotension_condition_present**
Orthostatic hypotension,[A][B] as indicated by 1 or more of the following(1)(2)(3):
 - **orthostatic_SBP_drop >= 20 mmHg**
Fall in SBP of 20 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position
 - **orthostatic_DBP_drop >= 10 mmHg**
Fall in DBP of 10 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position
- **persistence_despite_appropriate_treatment_context_present**
persists despite appropriate treatment
- **OR severe_adult_hypotension_condition_present**
Severe hypotension in adult patient, as indicated by 1 or more of the following:
 - **SBP < 80 mmHg**
Systolic blood pressure less than 80 mm Hg
 - **shock_index > 1.0 ratio**
Shock index greater than 1.0 (shock index is the heart rate divided by systolic blood pressure)
 - **MAP < 65 mmHg**
Mean arterial pressure[B] less than 65 mm Hg MAP Calculator[A](1)(6)
 - **vasopressor_or_inotrope_required**
IV inotropic or vasopressor medication required to maintain adequate blood pressure or perfusion
- **AND hypoperfusion_due_to_hypotension_condition_present**
Hypoperfusion due to hypotension, as indicated by ALL of the following:
 - **OR hypotension_condition_present**
Hypotension, as indicated by 1 or more of the following:
 - **OR hypotension_in_adult_patient_condition_present**
Hypotension in adult patient, as indicated by 1 or more of the following:
 - **SBP < 90 mmHg**
Systolic blood pressure less than 90 mm Hg[A][B](1)
 - **MAP < 70 mmHg**
Mean arterial pressure[C] less than 70 mm Hg[A][B] MAP Calculator(1)(2)
 - **baseline_sbp_decrease_with_symptoms_condition_present**
Decrease in baseline systolic blood pressure of 30 mm Hg or more, with significant signs or symptoms due to lower blood pressure (eg, near syncope, syncope, chest pain)[A][B](1)
 - **OR hypotension_in_pediatric_patient_condition_present**
Hypotension in pediatric patient, as indicated by 1 or more of the following:
 - **SBP < 110 mmHg (needs_review)**
Systolic blood pressure less than 110 mm Hg in child 13 to 17 years of age[A][D](3)
 - **SBP < 100 mmHg (needs_review)**
Systolic blood pressure less than 100 mm Hg in child 6 to 12 years of age[A][D](3)
 - **SBP < 95 mmHg (needs_review)**
Systolic blood pressure less than 95 mm Hg in child 3 to 5 years of age[A][D](3)
 - **SBP < 90 mmHg (needs_review)**
Systolic blood pressure less than 90 mm Hg in child 1 or 2 years of age[A][D](3)
 - **SBP < 80 mmHg (needs_review)**
Systolic blood pressure less than 80 mm Hg in infant 6 to 11 months of age[A][D](3)
 - **SBP < 70 mmHg (needs_review)**
Systolic blood pressure less than 70 mm Hg in infant 3 to 5 months of age[A][D](3)
 - **SBP < 65 mmHg (needs_review)**
Systolic blood pressure less than 65 mm Hg in infant 1 or 2 months of age[A][D](3)
 - **OR hypoperfusion_evidence_condition_present**
Evidence of hypoperfusion, as indicated by 1 or more of the following:
 - **lactate >= 2.0 mmol/L**
Lactate of 2.0 mmol/L (18 mg/dL) or more secondary to hypotension (ie, hypoperfusion)
 - **pH < 7.35 pH**
Metabolic acidosis (arterial or venous pH less than 7.35) not otherwise explained
 - **end_organ_dysfunction_due_to_hypotension_condition_present**
Significant end organ dysfunction due to hypotension (eg, Altered mental status, myocardial ischemia, 2-fold or higher acute increase in serum creatinine)
- **OR severe_pediatric_hypotension_condition_present**
Severe hypotension in pediatric patient, as indicated by 1 or more of the following:
 - **pediatric_systolic_blood_pressure_threshold_condition_present (needs_review)**
Systolic blood pressure less than 80 mm Hg in child 6 to 17 years of age[A](14) Systolic blood pressure less than 75 mm Hg in child 6 months to 5 years of age[A](14)
 - **pediatric_shock_index_threshold_condition_present (needs_review)**
Shock index greater than 0.9 in child 13 to 17 years of age (shock index is the heart rate divided by systolic blood pressure)[A](2)(15) Shock index greater than 1.0 in child 7 to 12 years of age (shock index is the heart rate divided by systolic blood pressure)[A](2)(15)
(15) Shock index greater than 1.2 in child 4 to 6 years of age (shock index is the heart rate divided by systolic blood pressure)[A](2)(15)

- **pediatric_vasopressor_or_inotrope_required (needs_review)**

IV inotropic or vasopressor medication required to maintain adequate blood pressure or perfusion

- **pediatric_hypoperfusion_due_to_hypotension_condition_present (needs_review)**

Hypoperfusion due to hypotension, as indicated by ALL of the following: Hypotension Evidence of hypoperfusion, as indicated by 1 or more of the following: Lactate of 2.0 mmol/L (18 mg/dL) or more secondary to hypotension (ie, hypoperfusion)[C](6)(8) Metabolic acidosis (arterial or venous pH less than 7.35) not otherwise explained[D](5)(6)(14) Significant end organ dysfunction due to hypotension (eg, Altered mental status, myocardial ischemia, 2-fold or higher acute increase in serum creatinine)(5)(6)(14)

Path 2: Severe electrolyte abnormalities requiring inpatient care))

- [atomic / IS_TRUE / severe_electrolyte_abnormality_requiring_inpatient_care_condition_present] Severe electrolyte abnormalities requiring inpatient care (3) (7)

Linked shared definition: [severe_electrolyte_abnormality_requiring_inpatient_care_condition_present](#) -> [comp.severe_electrolyte_inpatient.root](#)

Source: Definition - Severe electrolyte abnormalities requiring inpatient care

- **AND severe_electrolyte_abnormality_requiring_inpatient_care_condition_present (needs_review)**

Severe electrolyte abnormalities requiring inpatient care (ALL prerequisites + severity OR)

- **electrolyte_abnormality_not_at_acceptable_baseline_condition_present**

Electrolyte abnormality is not at acceptable patient baseline (eg, treatment effect).

- **OR severe_electrolyte_lab_criteria_or_branch_condition_present (needs_review)**

Severe abnormality, as indicated by 1 or more of the following:

- **severe_electrolyte_lab_line_candidate_condition_present (needs_review)**

Sodium[A] less than 130 mEq/L (mmol/L) not corrected to near normal or near chronic baseline despite observation care treatment(4)(5)

- **severe_electrolyte_lab_line_candidate_condition_present (needs_review)**

Sodium[A] less than 135 mEq/L (mmol/L) with new (or presumed new) severe finding requiring inpatient management (eg, Altered mental status, seizures)(4)(5)

- **severe_electrolyte_lab_line_candidate_condition_present (needs_review)**

Sodium greater than 150 mEq/L (mmol/L) not corrected to near normal or near chronic baseline despite observation care treatment(4)

- **severe_electrolyte_lab_line_candidate_condition_present (needs_review)**

Sodium greater than 145 mEq/L (mmol/L) with new (or presumed new) severe finding requiring inpatient management (eg, Altered mental status, seizures)(4)

- **severe_electrolyte_lab_line_candidate_condition_present (needs_review)**

Potassium less than 2.5 mEq/L (mmol/L) not corrected to near normal or near chronic baseline despite observation care treatment(6)

- **severe_electrolyte_lab_line_candidate_condition_present (needs_review)**

Potassium less than 3 mEq/L (mmol/L) with new (or presumed new) severe finding requiring inpatient management (eg, ECG findings,[B] paresis, paralysis)(6)(7)

- **severe_electrolyte_lab_line_candidate_condition_present (needs_review)**

Potassium greater than 6.5 mEq/L (mmol/L) not corrected to near normal or near chronic baseline despite observation care treatment(6)

- **severe_electrolyte_lab_line_candidate_condition_present (needs_review)**

Potassium greater than 5 mEq/L (mmol/L) with new (or presumed new) severe finding requiring inpatient management (eg, ECG findings,[C] paresis, paralysis)(6)(7)(8)

- **severe_electrolyte_lab_line_candidate_condition_present (needs_review)**

Calcium[D] less than 7 mg/dL (1.75 mmol/L) or ionized calcium less than 3.2 mg/dL (0.8 mmol/L) not corrected to near normal or near chronic baseline despite observation care treatment(9)(11)

- **severe_electrolyte_lab_line_candidate_condition_present (needs_review)**

Calcium[D] less than 8.6 mg/dL (2.15 mmol/L) or ionized calcium less than 4.65 mg/dL (1.16 mmol/L) with new (or presumed new) severe finding requiring inpatient management (eg, Altered mental status, seizures, arrhythmia, cardiac conduction disturbance)(9)(11)

- **severe_electrolyte_lab_line_candidate_condition_present (needs_review)**

Calcium[D] greater than 14 mg/dL (3.5 mmol/L) or ionized calcium greater than 10.0 mg/dL (2.50 mmol/L) not corrected to near normal or near chronic baseline despite observation care treatment(10)(11)

- **severe_electrolyte_lab_line_candidate_condition_present (needs_review)**

Calcium[D] greater than 12 mg/dL (3 mmol/L) or ionized calcium greater than 8.0 mg/dL (2.0 mmol/L) with new (or presumed new) severe finding requiring inpatient management (eg, Altered mental status, arrhythmia, cardiac conduction disturbance)(10)(11)

- **severe_electrolyte_lab_line_candidate_condition_present (needs_review)**

Phosphorus less than 1 mg/dL (0.32 mmol/L) not corrected to near normal or near chronic baseline despite observation care treatment(12)

- **severe_electrolyte_lab_line_candidate_condition_present (needs_review)**

Phosphorus less than 1.5 mg/dL (0.48 mmol/L) with new (or presumed new) severe finding requiring inpatient management (eg, Altered mental status, seizures)(12)

- **severe_electrolyte_lab_line_candidate_condition_present (needs_review)**

Phosphorus greater than 10 mg/dL (3.2 mmol/L) not corrected to near normal or near chronic baseline despite observation care treatment(12)

- **severe_electrolyte_lab_line_candidate_condition_present (needs_review)**

Phosphorus greater than 4.5 mg/dL (1.45 mmol/L) with new (or presumed new) severe finding requiring inpatient management (eg, crush injury, Altered mental status, seizures, arrhythmia, cardiac conduction disturbance)(12)

- **severe_electrolyte_lab_line_candidate_condition_present (needs_review)**

Magnesium less than 1 mg/dL (0.41 mmol/L) not corrected to near normal or near chronic baseline despite observation care treatment(12)

- **severe_electrolyte_lab_line_candidate_condition_present (needs_review)**

Magnesium less than 1.5 mg/dL (0.62 mmol/L) with new (or presumed new) severe finding requiring inpatient management (eg, Altered mental status, seizures, arrhythmia, cardiac conduction disturbance)(12)

- **severe_electrolyte_lab_line_candidate_condition_present (needs_review)**

Magnesium greater than 4.9 mg/dL (2 mmol/L) not corrected to near normal or near chronic baseline despite observation care treatment(12)

- **severe_electrolyte_lab_line_candidate_condition_present (needs_review)**

Magnesium greater than 3.0 mg/dL (1.23 mmol/L) with new (or presumed new) severe finding requiring inpatient management (eg, Altered mental status, arrhythmia, cardiac conduction disturbance)(12)

- **severe_electrolyte_lab_line_candidate_condition_present (needs_review)**

Uric acid greater than 20 mg/dL (1.19 mmol/L) not corrected to near normal or near chronic baseline despite observation care treatment(13)

- **severe_electrolyte_lab_line_candidate_condition_present (needs_review)**

Uric acid greater than 8 mg/dL (0.48 mmol/L) with new (or presumed new) severe finding requiring inpatient management (eg, arrhythmia, cardiac conduction disturbance, seizure)(13)

Path 3: Cardiac arrhythmias of immediate concern))

- [atomic / IS_TRUE / cardiac_arrhythmia_of_immediate_concern_condition_present] Cardiac arrhythmias of immediate concern (3) (7)

Linked shared definition: cardiac_arrhythmia_of_immediate_concern_condition_present -> comp.cardiac_arrhythmia_immediate.root

Source: Definition - Cardiac arrhythmias of immediate concern

- **OR cardiac_arrhythmia_of_immediate_concern_condition_present**

Cardiac arrhythmias or findings of immediate concern, as indicated by 1 or more of the following(1)(2)(3):

- **clinically_active_atrial_fibrillation_condition_present**

Clinically active atrial fibrillation refers to atrial fibrillation necessitating electrical cardioversion or treatment with parenteral (IV) medication, while hospitalized, to control heart rate (eg, digoxin, diltiazem, metoprolol) or to cardiovert (eg, amiodarone, procainamide).

- **OR other_immediate_cardiac_arrhythmia_concern_condition_present**

Any heart rhythm and 1 or more of the following(5)(8)(10)(11)(12)(13):

- **suspected_cardiac_ischemia_as_cause_or_consequence_of_vt_condition_present**

Suspected cardiac ischemia as cause or consequence of ventricular tachycardia

- **continuous_long_term_ecg_monitoring_needed_condition_present**

Continuous long-term ECG monitoring needed (eg, initiation of drug requiring monitoring beyond observation care)

- **automatic_icd_firing_malfunction_or_setting_adjustment_needed_condition_present**

Patient has automatic implantable cardioverter-defibrillator that is repeatedly firing, malfunctioning, or in need of immediate adjustment of settings beyond scope of observation care.

Path 4: Acute myocardial ischemia causing or associated with failure. See Angina ISC or Myocardial Infarction ISC guideline as appropriate.)))

- [atomic / IS_TRUE / acute_myocardial_ischemia_with_heart_failure_admission_condition_present] Acute myocardial ischemia causing or associated with failure. See Angina ISC or Myocardial Infarction ISC guideline as appropriate. (1) (3) (7)

Linked shared definition: acute_myocardial_ischemia_with_heart_failure_admission_condition_present -> comp.m190 verbatim.a7fe7d0528398c

Source: M190 Admission (synthetic linkage)

- **OR acute_myocardial_ischemia_with_heart_failure_admission_condition_present (needs_review)**

Acute myocardial ischemia causing or associated with failure (see Angina ISC or Myocardial Infarction ISC guideline as appropriate).

- **acute_myocardial_ischemia_with_heart_failure_admission_condition_present_verbatim_fact_condition_present (needs_review)**

Acute myocardial ischemia causing or associated with failure (see Angina ISC or Myocardial Infarction ISC guideline as appropriate).

Path 5: Pulmonary edema that is very severe (eg, invasive or noninvasive assisted ventilation needed, imminent, or likely; or need for 100% oxygen to keep oxygen saturation above 90%)))

- [atomic / IS_TRUE / very_severe_pulmonary_edema_condition_present] Pulmonary edema that is very severe (eg, invasive or noninvasive assisted ventilation needed, [A] imminent, or likely; or need for 100% oxygen to keep oxygen saturation above 90%) (3) (10)

Linked shared definition: very_severe_pulmonary_edema_condition_present -> comp.m190 verbatim.75fecdcfed3e65

Source: Definition - Respiratory distress

- **OR very_severe_pulmonary_edema_condition_present (needs_review)**

Pulmonary edema that is very severe (eg, invasive or noninvasive assisted ventilation needed, imminent or likely; or need for 100% oxygen to keep oxygen saturation above 90%).

- **very_severe_pulmonary_edema_condition_present_verbatim_fact_condition_present (needs_review)**

Pulmonary edema that is very severe (eg, invasive or noninvasive assisted ventilation needed, imminent or likely; or need for 100% oxygen to keep oxygen saturation above 90%).

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Path 6: Anasarca or peripheral edema that is severe (eg, significant paroxysmal dyspnea, inability to ambulate or void, skin breakdown, concomitant ascites)))

- [atomic / IS_TRUE / severe_anasarca_or_peripheral_edema_condition_present] Anasarca or peripheral edema that is severe (eg, significant paroxysmal dyspnea, inability to ambulate or void, skin breakdown, concomitant ascites) (2) (11)

Linked shared definition: severe_anasarca_or_peripheral_edema_condition_present -> comp.anasarca_severe_peripheral.root

Source: Definition - Anasarca

- **AND severe_anasarca_or_peripheral_edema_condition_present**

Severe anasarca/peripheral edema bundle (definition + admission clause).

- **anasarca_definition_generalized_edema_condition_present**

Anasarca is generalized edema that is widespread and may be more pronounced in dependent areas (eg, back while lying down, legs while sitting up).(1)

- **severe_peripheral_edema_manifestation_clause_condition_present**

Anasarca or peripheral edema that is severe (eg, significant paroxysmal dyspnea, inability to ambulate or void, skin breakdown, concomitant ascites)

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Path 7: Pulmonary edema and ALL of the following))

- [composite / AND / pulmonary_edema_with_oxygen_need_and_observation_failure_condition_present] Pulmonary edema and ALL of the following (3) (7):

Linked shared definition: pulmonary_edema_with_oxygen_need_and_observation_failure_condition_present -> comp.pe_with_o2_observation_bundle.root

Source: Definition - Hypoxemia

- **AND pulmonary_edema_with_oxygen_need_and_observation_failure_condition_present (needs_review)**

Pulmonary edema and ALL of the following (compound admission bullets).

- **pulmonary_edema_new_or_increased_oxygen_need_for_spo2_target_condition_present (needs_review)**

New need for oxygen therapy to keep oxygen saturation above 90% (or increased FiO2 need from baseline).

- **pulmonary_edema_insufficient_improvement_with_observation_care_diuretics_condition_present (needs_review)**

Has not improved sufficiently with observation care (eg, appropriately dosed IV diuretics).

- [atomic / IS_TRUE / leaf_m190_9efb819860096e4e] New need for oxygen therapy to keep oxygen saturation above 90% (or increased FiO 2 need from baseline)

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_27e5d3655fec392a] Has not improved sufficiently with observation care (eg, appropriately dosed IV diuretics) [B]

No shared definition linked yet - treated as current direct leaf / extractor target.

Path 8: Tachypnea that persists despite observation care)

- [atomic / IS_TRUE / persistent_tachypnea_despite_observation_care_condition_present] Tachypnea that persists despite observation care (7)

Linked shared definition: persistent_tachypnea_despite_observation_care_condition_present -> comp.persistent_tachypnea_observation.root

Source: Definition - Tachypnea

- **AND persistent_tachypnea_despite_observation_care_condition_present**

Definition - Tachypnea + admission persistence-after-observation clause.

- **OR tachypnea_condition_present**

Tachypnea,[A][B] as indicated by 1 or more of the following:

- **respiratory_rate > 20 breaths/min**

Respiratory rate greater than 20 breaths per minute in adult[A][B](1)

- **respiratory_rate > 18 breaths/min (needs_review)**

Respiratory rate greater than 18 breaths per minute in child 13 to 17 years of age[A][B](2)

- **respiratory_rate > 22 breaths/min (needs_review)**

Respiratory rate greater than 22 breaths per minute in child 6 to 12 years of age[A][B](2)

- **respiratory_rate > 25 breaths/min (needs_review)**

Respiratory rate greater than 25 breaths per minute in child 3 to 5 years of age[A][B](2)

- **respiratory_rate > 30 breaths/min (needs_review)**

Respiratory rate greater than 30 breaths per minute in child 1 or 2 years of age[A][B](2)

- **respiratory_rate > 40 breaths/min (needs_review)**

Respiratory rate greater than 40 breaths per minute in infant 6 to 11 months of age[A][B](2)

- **respiratory_rate > 45 breaths/min (needs_review)**

Respiratory rate greater than 45 breaths per minute in infant 3 to 5 months of age[A][B](2)

- **respiratory_rate > 55 breaths/min (needs_review)**

Respiratory rate greater than 55 breaths per minute in infant 1 or 2 months of age[A][B](2)

- **tachypnea_persistence_despite_observation_care_condition_present**

Tachypnea that persists despite observation care.

Path 9: Dyspnea (above baseline) that persists despite observation care)

- [atomic / IS_TRUE / persistent_dyspnea_above_baseline_despite_observation_care_condition_present] Dyspnea (above baseline) that persists despite observation care (7)

Linked shared definition: persistent_dyspnea_above_baseline_despite_observation_care_condition_present -> comp.dyspnea_persistent_placeholder.root

Source: Definition - Respiratory distress

- **OR persistent_dyspnea_above_baseline_despite_observation_care_condition_present (needs_review)**

Dyspnea (above baseline) that persists despite observation care

- **persistent_dyspnea_above_baseline_admission_clause_condition_present (needs_review)**

Dyspnea (above baseline) that persists despite observation care (admission bullet; fuller respiratory-distress decomposition deferred).

Path 10: Altered mental status that is severe or persistent)

- [atomic / IS_TRUE / altered_mental_status_severe_or_persistent_condition_present] Altered mental status that is severe or persistent (8)

Linked shared definition: altered_mental_status_severe_or_persistent_condition_present -> comp.altered_mental_status.severe_persistent.root

Source: Definition - Altered mental status that is severe or persistent

- **OR altered_mental_status_severe_or_persistent_condition_present**

Altered mental status (ie, different from baseline) that is severe or persistent, as indicated by 1 or more of the following(1)(2)(3)(4):

- **altered_mental_status_severe_persistent_clause_condition_present**
Confusional state (eg, disorientation, difficulty following commands, deficit in attention) that persists despite appropriate treatment (eg, of underlying cause, despite observation care)
- **altered_mental_status_severe_persistent_clause_condition_present**
Lethargy (eg, awake or arousable, but with drowsiness; reduced awareness of self and environment) that persists despite appropriate treatment (eg, of underlying cause, despite observation care)
- **altered_mental_status_severe_persistent_clause_condition_present**
Obtundation (ie, arousable only with strong stimuli, lessened interest in environment, slowed responses to stimulation)
- **altered_mental_status_severe_persistent_clause_condition_present**
Stupor (ie, may be arousable but patient does not return to normal baseline level of awareness)
- **altered_mental_status_severe_persistent_clause_condition_present**
Coma (ie, not arousable)

Path 11: Increased creatinine with reduction of more than 50% in estimated glomerular filtration rate from baseline eGFR - Adult Calculator)

- [atomic / IS_TRUE / acute_renal_insufficiency_evidence_gt_50pct_egfr_reduction_from_baseline_condition_present] Increased creatinine with reduction of more than 50% in estimated glomerular filtration rate from baseline [C] eGFR – Adult Calculator (3)

Linked shared definition: acute_renal_insufficiency_evidence_gt_50pct_egfr_reduction_from_baseline_condition_present -> comp.m190 verbatim.eb27736c36bed4

Source: Definition - Severe renal failure

- **OR acute_renal_insufficiency_evidence_gt_50pct_egfr_reduction_from_baseline_condition_present (needs_review)**

Increased creatinine with reduction of more than 50% in estimated glomerular filtration rate from baseline.

- **acute_renal_insufficiency_evidence_gt_50pct_egfr_reduction_from_baseline_condition_present_verbatim_fact_condition_present (needs_review)**
Increased creatinine with reduction of more than 50% in estimated glomerular filtration rate from baseline.

Path 12: Progressively (ongoing) rising creatinine with reduction of more than 25% in estimated glomerular filtration rate from baseline eGFR - Adult Calculator))

- [atomic / IS_TRUE / progressive_renal_insufficiency_evidence_gt_25pct_egfr_reduction_condition_present] Progressively (ongoing) rising creatinine with reduction of more than 25% in estimated glomerular filtration rate from baseline [C] eGFR – Adult Calculator (3) (7)

Linked shared definition: progressive_renal_insufficiency_evidence_gt_25pct_egfr_reduction_condition_present -> comp.m190 verbatim.db2927ca9dc33f

Source: Definition - Severe renal failure

- **OR progressive_renal_insufficiency_evidence_gt_25pct_egfr_reduction_condition_present (needs_review)**

Progressively (ongoing) rising creatinine with reduction of more than 25% in estimated glomerular filtration rate from baseline.

- **progressive_renal_insufficiency_evidence_gt_25pct_egfr_reduction_condition_present_verbatim_fact_condition_present (needs_review)**
Progressively (ongoing) rising creatinine with reduction of more than 25% in estimated glomerular filtration rate from baseline.

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Path 13: Pulmonary artery catheter monitoring needed))

◦ [atomic / IS_TRUE / pulmonary_artery_catheter_monitoring_needed_condition_present] Pulmonary artery catheter monitoring needed [D] (1) (8)

Linked shared definition: pulmonary_artery_catheter_monitoring_needed_condition_present -> comp.m190 verbatim.fb1f0682700ae2

Source: Definition - Observation care

- **OR pulmonary_artery_catheter_monitoring_needed_condition_present** (needs_review)
Pulmonary artery catheter monitoring needed.
- **pulmonary_artery_catheter_monitoring_needed_condition_present_verbatim_fact_condition_present** (needs_review)
Pulmonary artery catheter monitoring needed.

M190 — Discharge

Discharge domain (planning + destination checklist items)

Section 1: Discharge planning includes

- [composite / CHECKLIST] Discharge planning includes [M] :
 - [composite / CHECKLIST] Assessment of needs and planning for care, including (46) (47) :
 - [atomic / IS_TRUE / leaf_m190_1da47a57bd1287a5] Develop and modify treatment plan (involving multiple providers) as needed.
No shared definition linked yet - treated as current direct leaf / extractor target.
 - [atomic / IS_TRUE / leaf_m190_901d0b649ea21dba] Evaluate and address preadmission functioning as needed.
No shared definition linked yet - treated as current direct leaf / extractor target.
 - [atomic / IS_TRUE / leaf_m190_f50c6dc1549c88b3] Evaluate and address psychosocial status issues as indicated. See Psychosocial Assessment SR for further information.
No shared definition linked yet - treated as current direct leaf / extractor target.
 - [atomic / IS_TRUE / leaf_m190_f3fc1a219ba383e7] Evaluate and address social determinants of health (eg, housing, food). See Social Determinants of Health Screening Tool SR for further information. (48) (49)
No shared definition linked yet - treated as current direct leaf / extractor target.
 - [atomic / IS_TRUE / leaf_m190_7b38af0f0d3e74ae] Evaluate and address patient or caregiver preferences as indicated.
No shared definition linked yet - treated as current direct leaf / extractor target.
 - [composite / CHECKLIST] Identify skilled services needed at next level of care, with specific attention to:
 - [atomic / IS_TRUE / leaf_m190_77bf61ecb1c8f6bd] Cardiovascular status assessment (50)
No shared definition linked yet - treated as current direct leaf / extractor target.
 - [atomic / IS_TRUE / leaf_m190_d62a340c381b9ec7] Medication management, adherence instruction, and side effects assessment (51)
No shared definition linked yet - treated as current direct leaf / extractor target.
 - [atomic / IS_TRUE / leaf_m190_13e9f47f92112a2e] Nutrition and hydration management (52) (53)
No shared definition linked yet - treated as current direct leaf / extractor target.
 - [atomic / IS_TRUE / leaf_m190_67247a9856e7573c] Ongoing education required (53)
No shared definition linked yet - treated as current direct leaf / extractor target.
 - [atomic / IS_TRUE / leaf_m190_291641275a8ec331] Psychosocial assessment, management, and referrals (54)
No shared definition linked yet - treated as current direct leaf / extractor target.
 - [composite / OPTIONS] Early identification of anticipated discharge destination; options include (47) (55) :
 - [composite / CHECKLIST] Home; considerations include:
 - [atomic / IS_TRUE / leaf_m190_d0b0fc566902b761] Home safety assessment. See Home Safety Assessment SR for further information.
No shared definition linked yet - treated as current direct leaf / extractor target.
 - [composite / EXAMPLE_SET] Patient safe to go home; examples include (56) (57) (58) :
 - [atomic / IS_TRUE / leaf_m190_eaa60c4689149326] Medical status stable for patient's condition
No shared definition linked yet - treated as current direct leaf / extractor target.
 - [atomic / IS_TRUE / leaf_m190_dcb685270158033d] Functional care can safely be provided with available resources.
No shared definition linked yet - treated as current direct leaf / extractor target.
 - [atomic / IS_TRUE / leaf_m190_d8d4b3d9ecb24006] Mental status stable for patient's condition
No shared definition linked yet - treated as current direct leaf / extractor target.
 - [atomic / IS_TRUE / leaf_m190_37ded66410b9898d] Medication availability confirmed and reconciliation complete
No shared definition linked yet - treated as current direct leaf / extractor target.
 - [atomic / IS_TRUE / leaf_m190_7e7d8622872c838c] Patient/caregiver education completed with written discharge instructions provided
No shared definition linked yet - treated as current direct leaf / extractor target.
 - [atomic / IS_TRUE / leaf_m190_6abd192c8b2e0559] Community resources identified and referrals made, as needed
No shared definition linked yet - treated as current direct leaf / extractor target.
 - [atomic / IS_TRUE / leaf_m190_ff4ce25f9e5277b5] Home care arranged, if indicated
No shared definition linked yet - treated as current direct leaf / extractor target.
 - [atomic / IS_TRUE / leaf_m190_70ce60c0c8c8d218] Necessary medical equipment delivery arranged or available in home, if indicated
No shared definition linked yet - treated as current direct leaf / extractor target.
 - [atomic / IS_TRUE / leaf_m190_94389b8025d0b448] Necessary medical supplies ordered, or patient/caregiver can obtain, if indicated
No shared definition linked yet - treated as current direct leaf / extractor target.
 - [atomic / IS_TRUE / leaf_m190_7d727e79f2d2d361] Access to follow-up care
No shared definition linked yet - treated as current direct leaf / extractor target.
 - [atomic / IS_TRUE / leaf_m190_9ccea61ea702694] Self-management ability if appropriate. See Activities of Daily Living (ADL) and Instrumental Activities of Daily Living (IADL) Assessment SR for further information.

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_4b2114703ffdb975] Caregiver need, ability, and availability

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_fa7c6751b8eebe59] Post-acute skilled care or custodial care as indicated. See Discharge Planning Tool SR for further information.

No shared definition linked yet - treated as current direct leaf / extractor target.

- [composite / CHECKLIST] Transitions of care plan complete, including (47) (55) (59):

- [composite / CHECKLIST] Patient and caregiver education complete.

- [atomic / IS_TRUE / leaf_m190_53431d91506d0e24] See Teach-Back Tool SR for further information.

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_d46f4ff2b6dfadeb] See Heart Failure: Patient Education for Clinicians SR for further information.

No shared definition linked yet - treated as current direct leaf / extractor target.

- [composite / CHECKLIST] Medication reconciliation completion includes (51) (60):

- [atomic / IS_TRUE / leaf_m190_1a045269ec0e95d2] Compare patient's discharge list of medications (prescribed and over-the-counter) against provider's admission or transfer orders.

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_91cdeee74f94d4c8] Assess each medication for correlation to disease state or medical condition.

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_409aaf63a315b9b0] Report medication discrepancies to prescribing provider, attending physician, and primary care provider, and ensure accurate medication order is identified.

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_a7a3298eee6b4063] Provide reconciled medication list to all treating providers.

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_292bfe0c6f25626a] Confirm that patient or caregiver can acquire medication.

No shared definition linked yet - treated as current direct leaf / extractor target.

- [composite / CHECKLIST] Educate patient and caregiver.

- [atomic / IS_TRUE / leaf_m190_b2c4bc1e0b005abb] Provide complete medication list to patient and caregiver.

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_b532e84f55aa047d] Importance of presenting personal medication list to all providers at each care transition, including all provider appointments

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_c3682455f885a089] Reason, dosage, and timing of medication (eg, use "teach-back" techniques) (61)

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_5a7a51adebe0d0ac] Encourage communication between patient, caregiver, and pharmacy for obtaining prescriptions, setting up home medication delivery, and reviewing for drug-drug interactions.

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_f102abf07440f2f5] See Medication Reconciliation Tool SR for further information.

No shared definition linked yet - treated as current direct leaf / extractor target.

- [composite / CHECKLIST] Plan communicated to patient, caregiver, and all members of care team, including (62):

- [atomic / IS_TRUE / leaf_m190_472800e932382808] Inpatient care and service providers

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_f227baca0faa5eb7] Primary care provider

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_4ed88bdb4ff2203e] All post-discharge care and service providers

No shared definition linked yet - treated as current direct leaf / extractor target.

- [composite / OPTIONS] Appointments planned or scheduled, which may include:

- [atomic / IS_TRUE / leaf_m190_8eb72e3ef08c3083] Primary care provider (52) (63) (64)

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_9a3cc4a8c786f581] Cardiologist (65)

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_e7b77f58e7acc234] Dietitian (52)

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_97cebee68ba7da6c] Follow-up evaluation call. See Post-Acute Care Follow-Up Call Tool SR for further information. (1) (2) (63) (66)

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_3874c684aba507e9] Palliative or end-of-life care provider (67)

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_d05f3a4a1cec69b9] Rehabilitation therapy services (2) (52)

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_c8ee48d9b17fc5fe] Specialists for management of comorbidities as needed (68)

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_027616981c0a2176] Other

No shared definition linked yet - treated as current direct leaf / extractor target.

- [composite / OPTIONS] Outpatient testing and procedure plans made, which may include:

- [atomic / IS_TRUE / leaf_m190_b753cba47111a8d1] Echocardiogram

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_a8523323cf71d054] Exercise stress test (52)

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_71a4c8fa3b35941b] Laboratory testing (69)

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_b935ab8b0a39a66e] Other

No shared definition linked yet - treated as current direct leaf / extractor target.

- [composite / OPTIONS] Referrals made for assistance or support, which may include:

- [atomic / IS_TRUE / leaf_m190_5d86f49c0a6c0be8] Alcohol and other drug abuse or dependence treatment (70)

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_60a16ec7ef92f490] Community services

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_bcc907b15ec67d02] Educational program (eg, chronic condition management, self-management) (64) (69) (71)

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_f06f88cc7e04bcc] End-stage disease planning (52)

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_fe5ef25618a521ae] Financial, for follow-up care, medication, and transportation

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_cf229d4f83f0c995] Self-help or support groups (52)

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_41b7f5e81fe4807b] Social services (eg, social programs, advance directives) (71)

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_2e597b3c84c5d8aa] Tobacco use treatment (52) (69) (70) (72)

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_916d077431b92785] Other

No shared definition linked yet - treated as current direct leaf / extractor target.

- [composite / OPTIONS] Medical equipment and supplies coordinated (ie, delivered or delivery confirmed), which may include (52):

- [atomic / IS_TRUE / leaf_m190_df9f6ef3bd09a8e8] Blood pressure cuff

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_86bb772a8f64594a] Weight scale (2)

No shared definition linked yet - treated as current direct leaf / extractor target.

- [atomic / IS_TRUE / leaf_m190_29b30a56675c4eeb] Other

No shared definition linked yet - treated as current direct leaf / extractor target.

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Section 2: Discharge Destination

- [composite / OPTIONS] Post-hospital levels of admission may include:
 - [atomic / IS_TRUE / leaf_m190_ec4ec62cc2d44058] Home.
No shared definition linked yet - treated as current direct leaf / extractor target.
 - [atomic / IS_TRUE / leaf_m190_2747153c99c4cadb] Home healthcare. See Home Care Indications for Admission Section HC in Heart Failure guideline in Home Care.
No shared definition linked yet - treated as current direct leaf / extractor target.
 - [atomic / IS_TRUE / leaf_m190_6e742c6d7bd9c58d] Recovery facility care. See Recovery Facility Care Indications for Admission Section RFC in Heart Failure guideline in Recovery Facility Care.
No shared definition linked yet - treated as current direct leaf / extractor target.
 - [atomic / IS_TRUE / leaf_m190_514f740362826ec9] Inpatient rehabilitation facility. See Inpatient Rehabilitation Facility (Acute Rehabilitation) Indications for Admission Section RFC in Inpatient Rehabilitation Facility (Acute Rehabilitation): Cardiac Disorders guideline in Recovery Facility Care.
No shared definition linked yet - treated as current direct leaf / extractor target.